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Method Validation

5.1 Review of Validation Procedures

For many decades, analysts applied analytical methods developed in their laboratories. They rarely fully documented the operating procedure, as well as the tips and tricks related to their own experience, and often preferred an oral transmission of their knowledge. Obviously, they knew that random and systematic errors due to matrix interferences or incorrect manipulations might occur, but they also believed that their expertise was good enough to detect them and guarantee the correct interpretation of the result. This state of mind ended in the fifties for the Pharmaceutical community with the tragic problem of Thalidomide and the mid-1970s when various industrial accidents occurred (Minamata, Bhopal, Seveso, etc.).

It led end-users, especially control authorities in charge of public health or the environment, comparing laboratory results with others, to surprisingly realize that extremely high differences could be observed. For instance, in the 1980s, differences as high as to 1/1000 could be observed when analyzing chlorinated pesticides in foods. Motivated by the expectations of their customers, analysts set about two complementary tasks: harmonize the operating procedures and validate the methods.

For the first goal, working groups were established within national and international standardization organizations, such as ISO, or professional structures, such as AOAC. This resulted in many documents being presented in various forms. It is beyond our scope to describe the work that has been done. However, the fundamental role of standardization and harmonization efforts in improving the metrological quality of methods must be underlined and encouraged.

Here again, it is often through collaborative work that the second objective was achieved. Two complementary tasks can be distinguished:

- Develop the vocabulary with definitions of various performance characteristics or figures of merit.
- Develop the validation procedures with respect to expected performance criteria.

Despite the generic aspect of this program – since all analytical methods require validation regardless of their field of application – it has been implemented by a wide variety of organizations, resulting in various proposals and sometimes confusion.

The emergence of the need for validation of analytical methods can be dated back to the late 1980s.

This period overlaps with the publication by ISO of the first international standards on quality assurance: the ISO 9000 series. In these documents, many concepts related to “quality and all operations that contribute to it” are defined. Validation can be considered as a procedure to reach the adequate quality level for an analytical measurement. In the meantime, it was classically claimed that the quality of products or services must be designed to “satisfying customer needs.” This is also known as a “consumer-driven quality,” which was summarized in a five-point list when applicable to the laboratory:

- Meet a well-defined need, use, or objective.
- Fulfill consumer expectations.
- Comply with standards, specifications, regulations, and other requirements.
- Have at a competitive price.
- Provide data at a cost that generates a profit.

Considering a more recent context, new requirements should be added:

- Reduce the environmental impact of laboratories.
- Avoid reagents and substances unsafe for personnel and the environment.

From the beginning, the application of quality assurance principles in the laboratory has been perceived as a specific objective that required as such, specific texts. Today, the list is long of guides, recommendations, and guidelines published by various regulatory or professional structures on the application of quality assurance in the laboratory and on method validation. For example, a large set of official documents available on the Internet describes all operations required to fulfill quality objectives [1–6]. Different comparative reviews of these documents have been published [7, 8].

Validation *per se* of methods often fills a short section of these documents, but it is always addressed. In most guides, the principle to conduct the validation consists in collecting experimental data that are processed to assess performance characteristics, such as accuracy, precision, limit of detection and limit of quantification, or sensitivity. Sometimes, schemes are also prescribed for the practical organization of the experiments.

But, except for some guidances, little or no advice is given on the number of measurements to be collected for a correct estimation of the performance characteristic nor on the experimental design to be applied. The next Sections 5.4.2 and 5.4.3 explain how the appropriate number of replicates is rarely used and may impact the reliability of obtained results and how a correct design of experiment can be crucial. Consequently, the decision rule for concluding whether a method is valid or not is not always very conclusive.

Over the last decades, validation has become an important concern for many analysts, and many of them have acquired obvious expertise. However, it may be useful to clarify some incoherent points, such as the vocabulary and the structure of the experimental design, where ambiguities persist. The imprecision of the classic

validation guides led us to develop a more structured approach. A statistical and graphical tool for method validation, called method accuracy profile (MAP), resulted from this observation.

A clear interpretation strategy is also in the pipeline. Over the past two decades, numerous papers using the accuracy profile have been published, as explained in this chapter. For several official standardization bodies, *validation* is synonymous with interlaboratory study. But, in many situations, method validation is internally conducted in a single laboratory and is sometimes called in-house validation. For example, if the method is:

- Developed for a time limited research topic.
- Applied to a small series of samples.
- Used to control the manufacture of a single product by a single producer.
- Transferred from another laboratory and its reliability needs to be verified.
- Restarted after a period of inactivity.
- The property of a laboratory that does not wish to transmit it to others.
- Multianalyte and multimatrix, for which an interlaboratory study is difficult to organize.

For most of these cases, it is not necessary or possible to involve a number of laboratories to validate a method. More generally, interlaboratory validation applies to a method in a pre-competitive context, such as health or official control, whereas an in-house validation applied to a method used in a competitive context, such as drug control.

5.1.1 Inconsistencies of Validation Vocabulary

While validation is a common requirement for all analysts for all methods, it is paradoxical that the documents related to this topic are generally presented as specific to an application sector such as drugs, foods, metals, medical biology, oenology, water, or mining. It is even more surprising because laboratories, whatever they work on, use the same analytical instruments and have the same validation needs. Among them, the main differences are:

- In the pre-analytical sampling step and sample preparation,
- In the final validation criterion.

As detailed in Section 8.4.3 about the definition of a replicate, only the instrumental step is purely *analytical* as it requires measuring principles based on chemistry or biology. The economic evolution of laboratories over the last few decades has pushed them to get together within larger industrial organizations and to be much more flexible than in the past.

For example, some years ago, soil and water analyses were conducted in specialized and independent laboratories. Today, it is common for these types of activities to be associated within a single legal structure and, sometimes, even in the same location. Because of the classic sectorial or *vertical* organization of many application fields, a diverse list of definitions has been independently issued, raising several communication problems.

To avoid some of these ambiguities, the reference proposed in this book is the International Vocabulary of Metrology or VIM, published by the Bureau International des Poids et Mesures or BIPM [9]. It is a well-accepted thesaurus applied in various fields of measurement. The definitions of many terms, when they exist, apply to analytical sciences, although some are missing or unsuitable. Most of the definitions are quoted in the text and written between double quotes. As a remainder, in VIM, two major vocabulary elements can be identified: characteristics and parameters.

Characteristics	They are <i>concepts</i> not “directly measurable” such as trueness, precision, or accuracy. For example, note 1, coming with the definition of measurement accuracy (VIM clause 2.13), states that: “measurement accuracy is not a quantity and is not expressed numerically. A measurement is sometimes said to be more accurate if it provides a smaller measurement error.”
Parameters	They are estimated from experimental data using a computational algorithm and called <i>estimators</i> by statisticians, i.e. a rule for calculating an estimate of a given quantity based on observed data. They are the coefficients of mathematical or statistical models.

For example note 1 on measurement precision (VIM clause 2.15) states that: “precision is usually expressed numerically by parameters such as standard deviation, variance or the coefficient of variation under the specified conditions.” If one parameter gives a numerical evaluation of a characteristic, then one characteristic can have several parameters alone or in combination. The numerical values of the parameters are sometimes referred to as *performance scores* or *figures-of-merit*.

A third category of vocabulary elements must be added, absent from the VIM, but essential to validate a method.

Criteria	They are <i>technical specifications</i> expressed as a minimum, maximum, acceptance interval, etc. They are often quantitative but sometimes qualitative, such as practicability or robustness. The procedural manual of the Codex Alimentarius Commission (CAC) [10] by means of the Codex Committee on Methods of Analysis and Sampling (CCMAS), proposes a Criteria Approach for selecting methods of analysis for food trade that is based on a set of criteria.
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A “Glossary of Used Terms” is proposed as an attempt to sum up some the usual definitions. It results from the compilation of various documents, including the VIM. It is not intended to be exhaustive or referential, as several instrumental methods or sectors of activity have developed their own vocabulary, but only to collate the definitions of the main terms used in this book.

Given the inconsistency of glossaries, it is not surprising that several authors underlined the divergences in published documents. The same term can have different definitions and meanings, sometimes difficult to relate to. The term “method validation” illustrates these discrepancies, as it can mean:

- “Verification, where the specified requirements are adequate for an intended use” (VIM), but also.
- Interlaboratory comparison that leads to precision parameters, such as reproducibility.

- Demonstration that the method measures the analyte it is supposed to. This definition is used in the title of many publications as a synonym for specificity.

The term “linearity” also has different meanings:

- Proportionality between the known contents of reference material and the inverse-predicted concentration, i.e. linearity of trueness.
- Existence of a linear relationship between instrumental response and analyte concentration, i.e. linearity of the calibration.

For the limit of quantification (LOQ), dozens of definitions are available, and even more calculation methods. Yet it is the most widely used parameter by analysts or equipment manufacturers to advertise the performance of an analytical technique. This topic is discussed in more detail in Section 9.1.

Another critical issue is the eventual confusion between the terms applicable to quantitative methods with those used for qualitative tests, as their concepts are different. A compilation of nearly forty validation guides, published mainly by international organizations, identified various inconsistencies. Five benchmarks were selected to conduct this textual analysis [8]. They are concerned with the type of analysis, chemical or biological, the national or international scope of the guide, the sector or discipline of origin, the analytical technique, and the compounds analyzed.

The various validation guides share a common set of terms, despite variable definitions, as shown in Figure 5.1. Perhaps it is this appearance of similarity that currently prevents a global standardization effort from taking shape.

5.1.2 Validation Plans

When scrutinizing literature, it is possible to identify a *classic validation procedure* that is widely published. IUPAC directive or ICH Q2(R2) guidelines are typical examples of such a procedure [11, 12]. The content of a classic validation procedure is as follows:

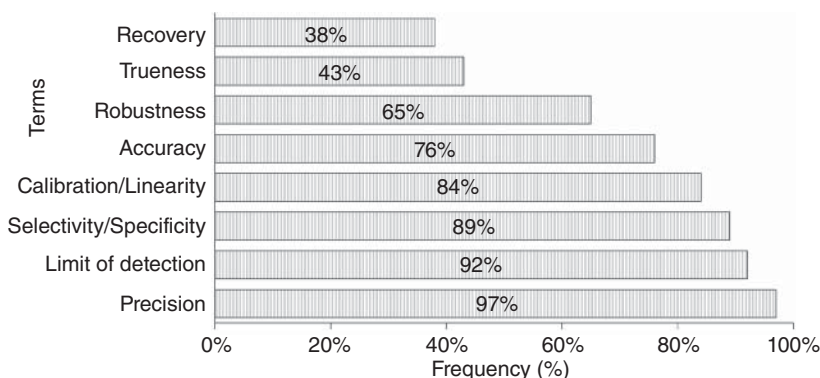


Figure 5.1 Frequency of terms used in validation guides. Source: Adapted from Raposo and Ibelli-Bianco [8].